Priority Queues and Heaps

Prof. Timothy Sun

Priority Queues

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How does one ***implement*** a priority queue?

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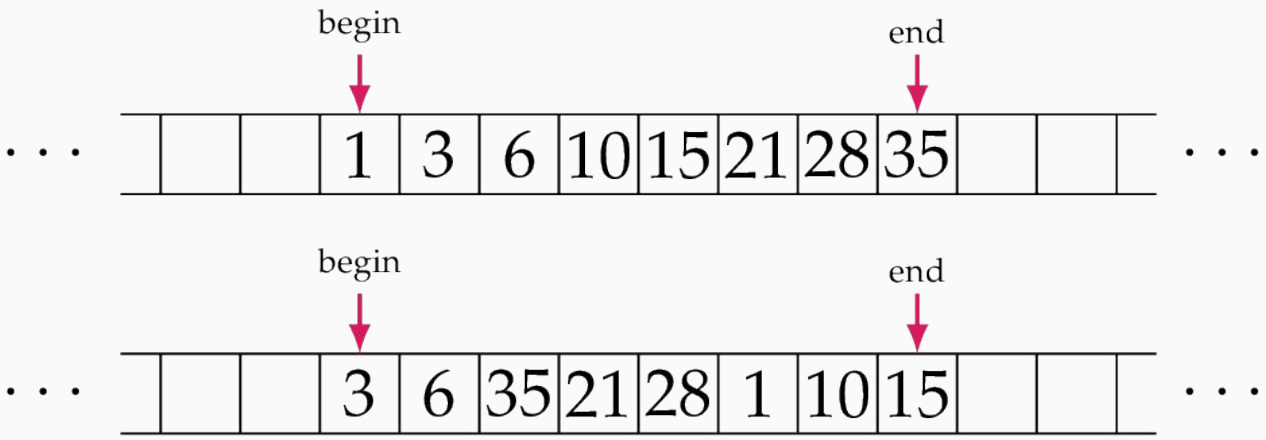


Sorted and unsorted implementations both seem plausible.

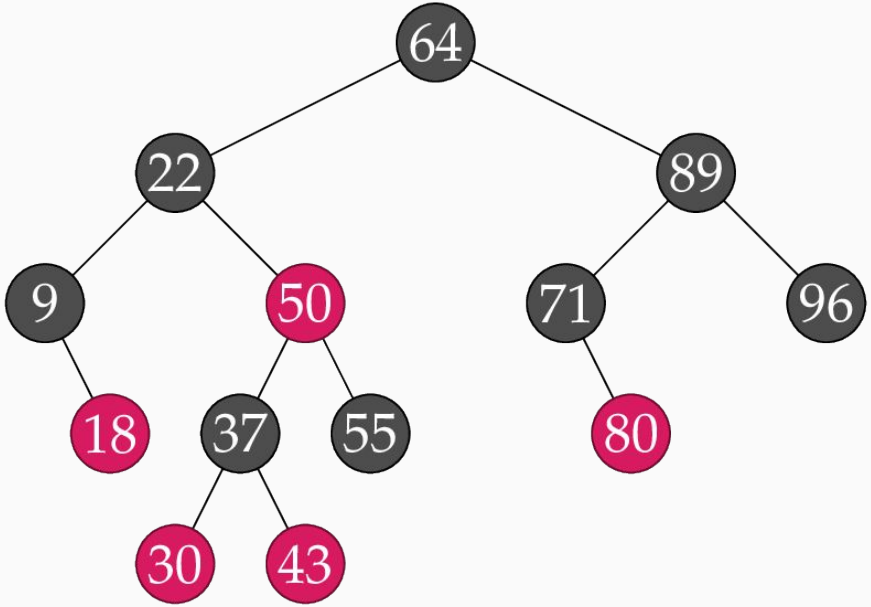
Array/List-based implementations

Each implementation has its strength, and a huge drawback:

	Unsorted List	Sorted List
insert	O(1)	**O(*n*)**
removeMin	**O(*n*)**	O(1)

Balanced BSTs?

One data structure that would solve this issue is a ***balanced binary search tree*** (e.g. AVL, red-black, etc.)



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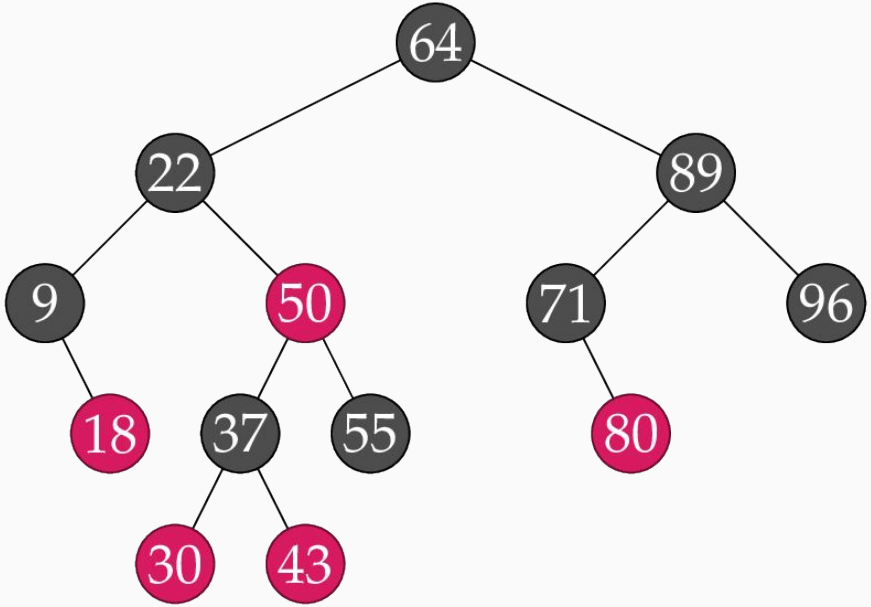

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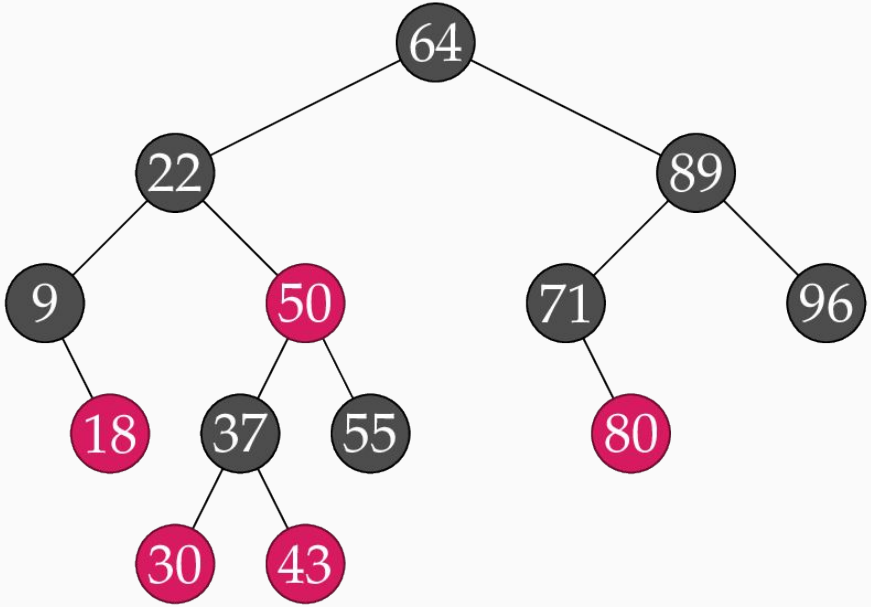

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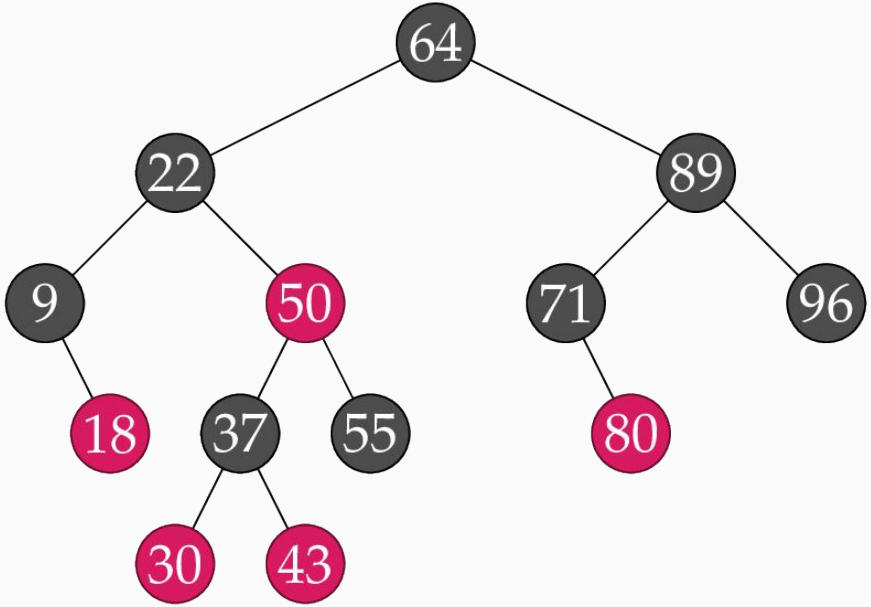
But this is a little overkill…



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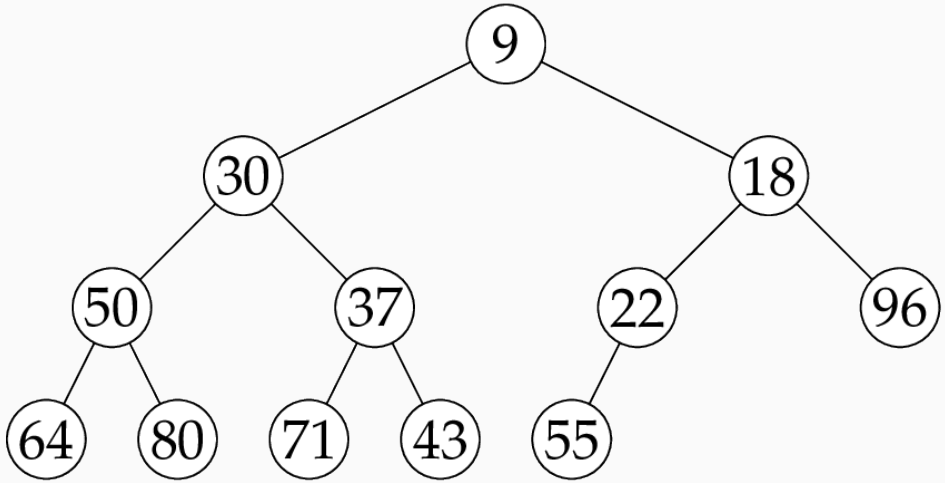

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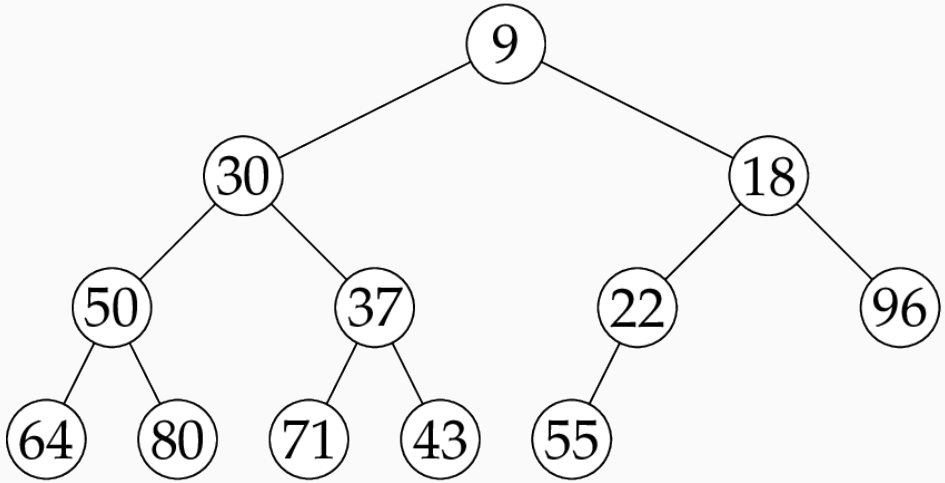
Two key properties:

- Heap order
- Complete binary tree



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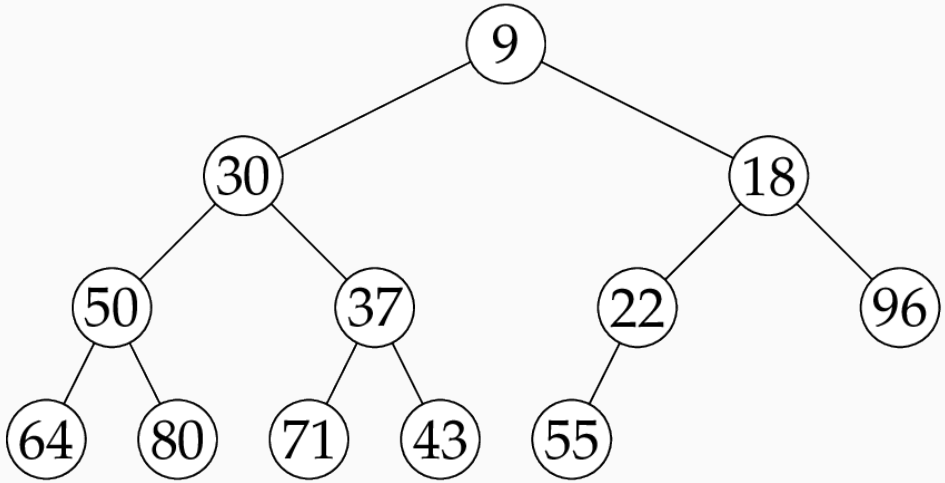
Heap order property: parent always smaller than child



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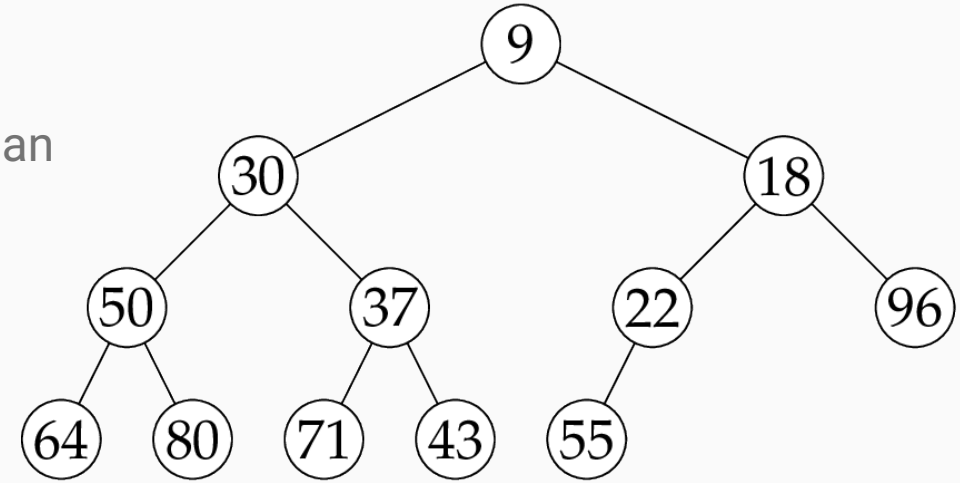

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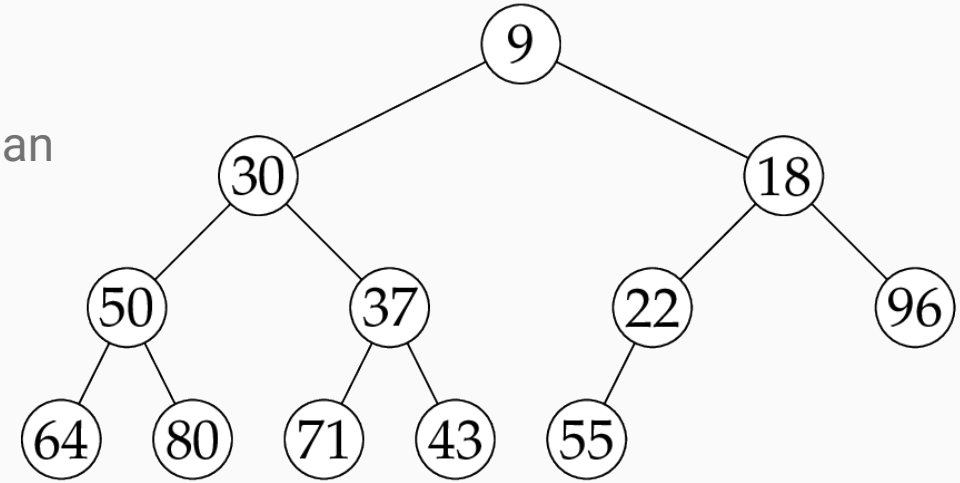

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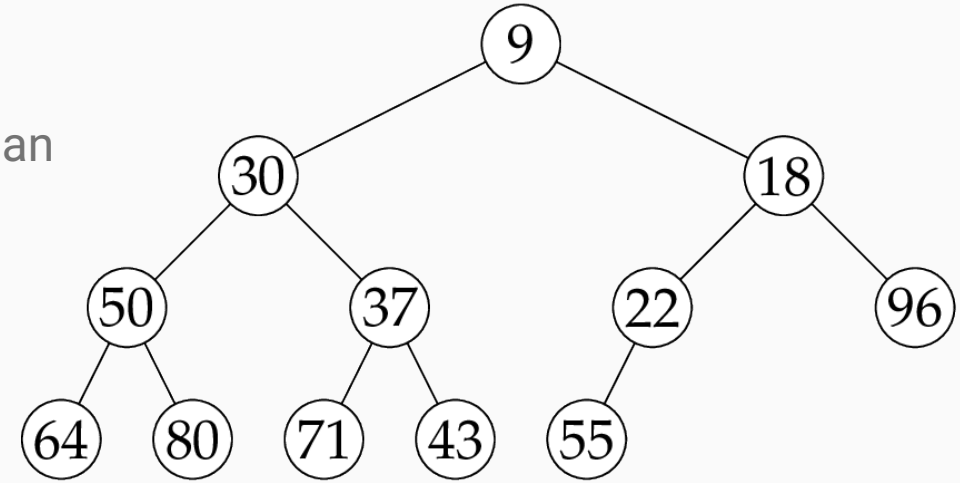
Where's the smallest element?

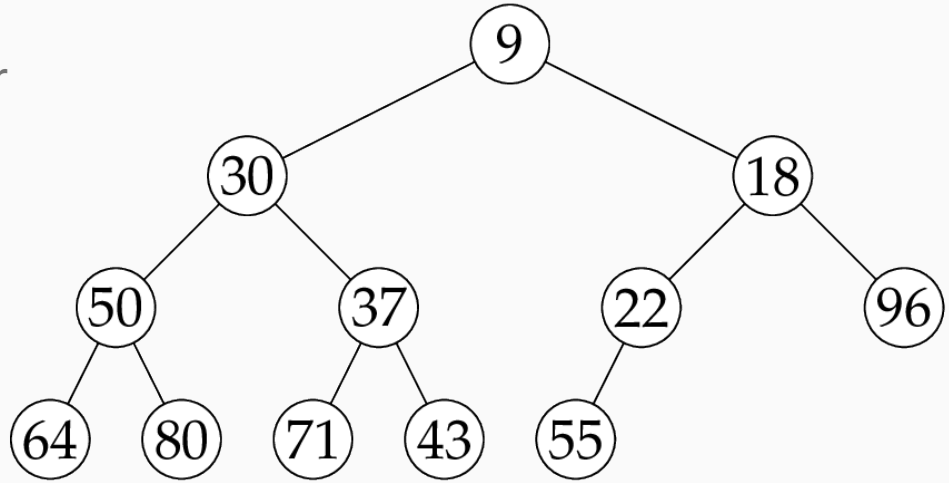
At the root.



Complete binary tree

Complete binary tree property:

- Except for last layer, every layer is “full”
- All the leaves appear as far left as possible



insert( ) - bubbling up

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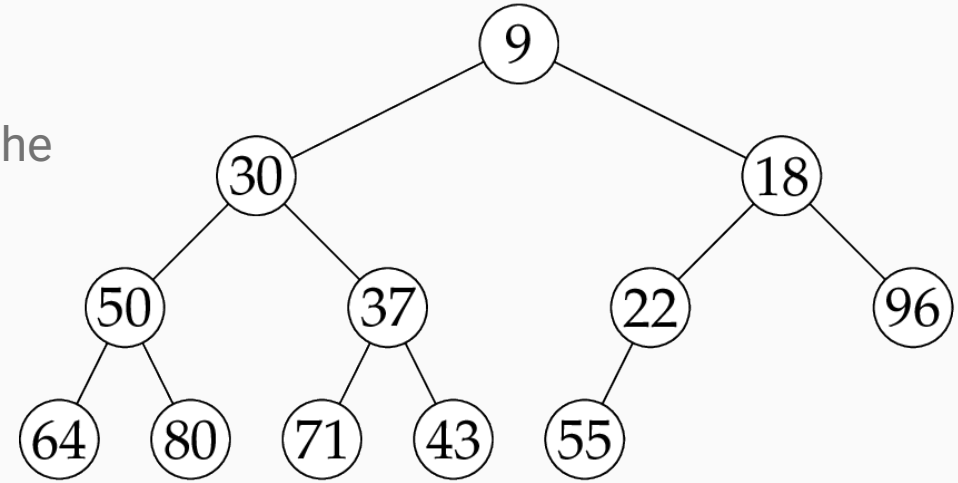
How to add a node that preserves the ***complete binary tree property***?



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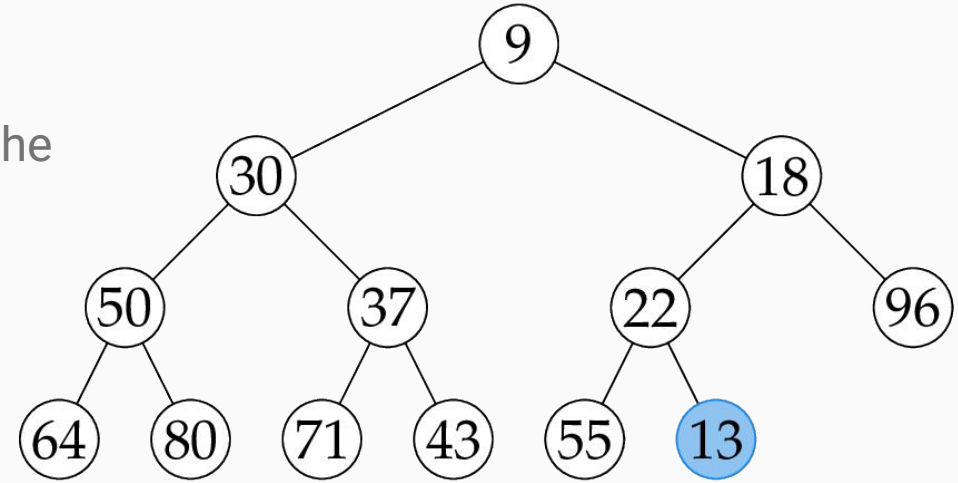

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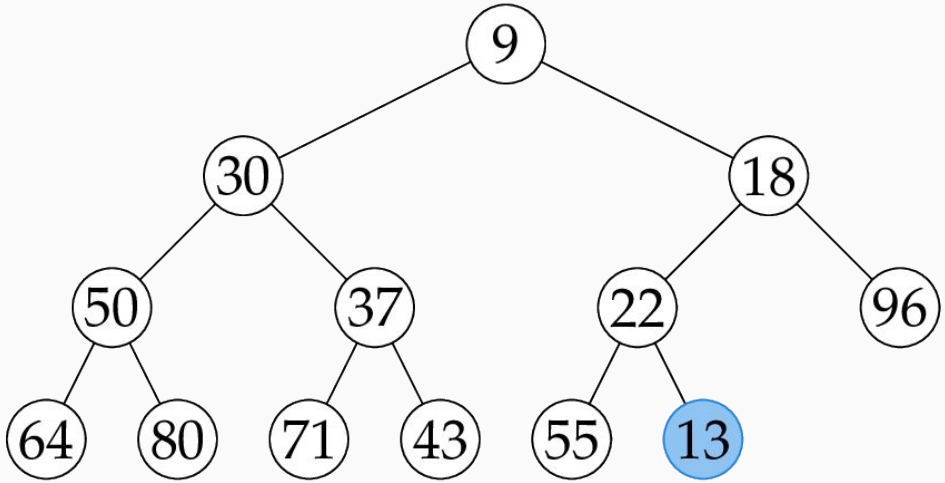

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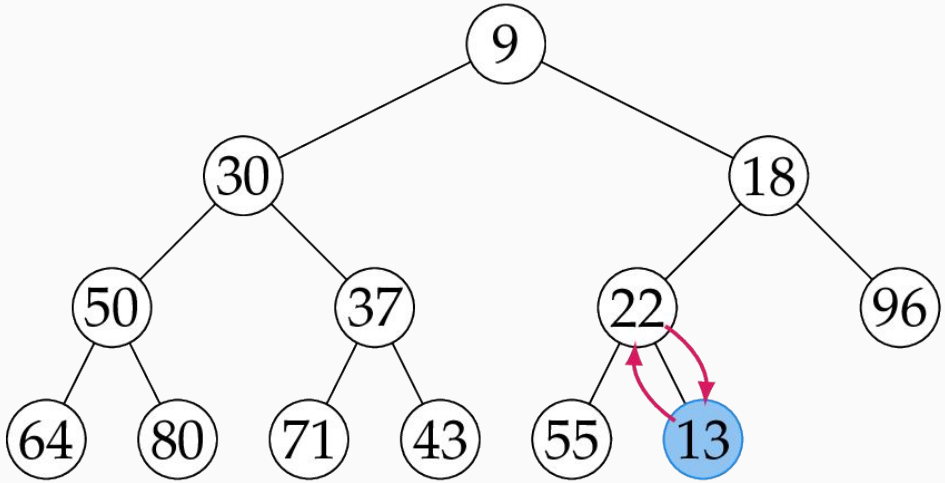

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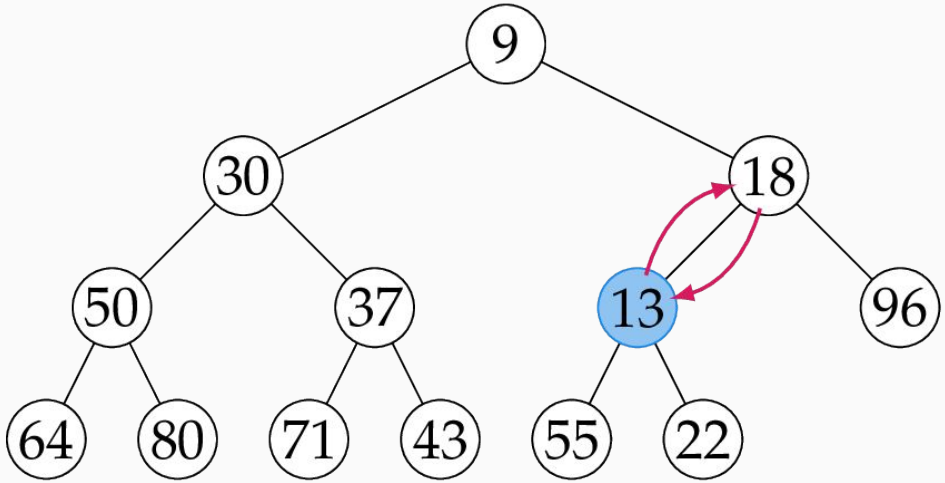

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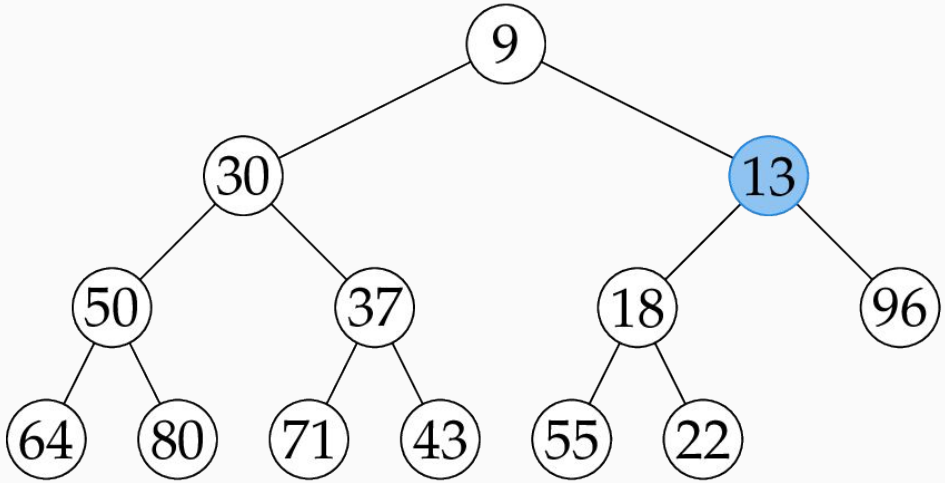

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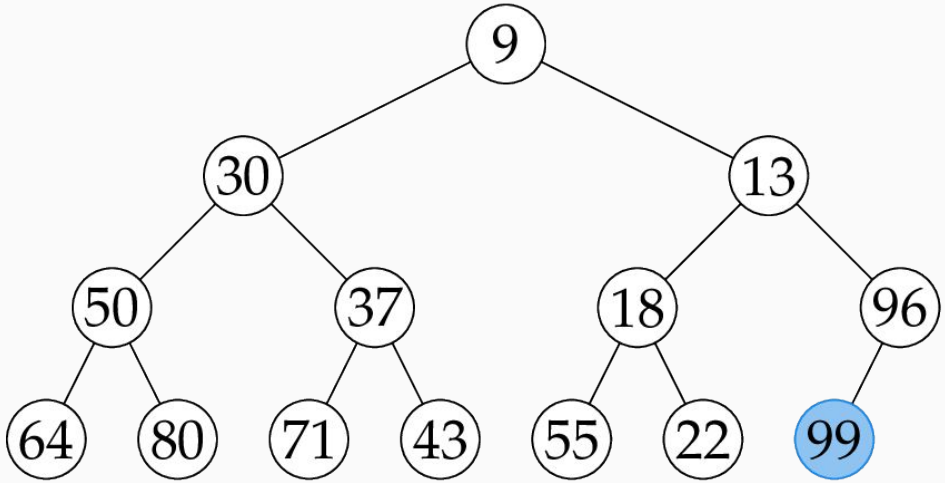

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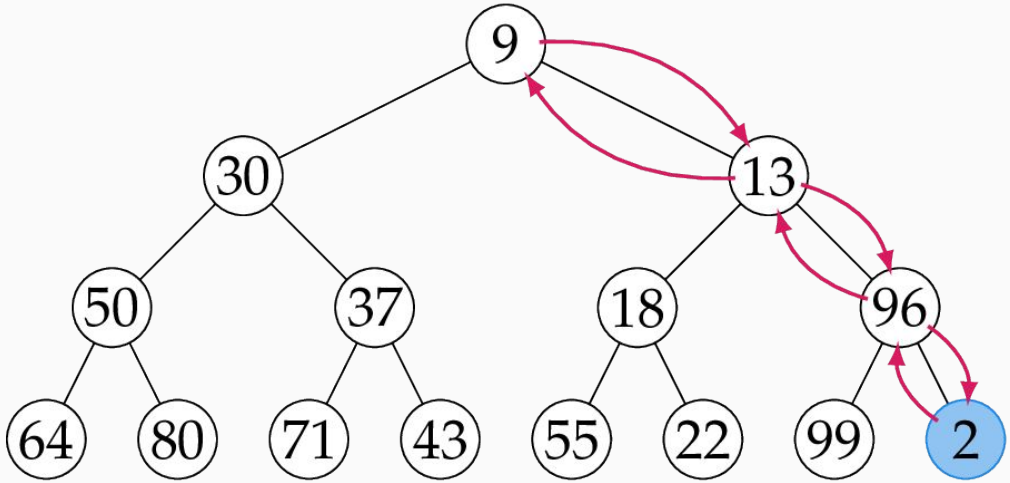

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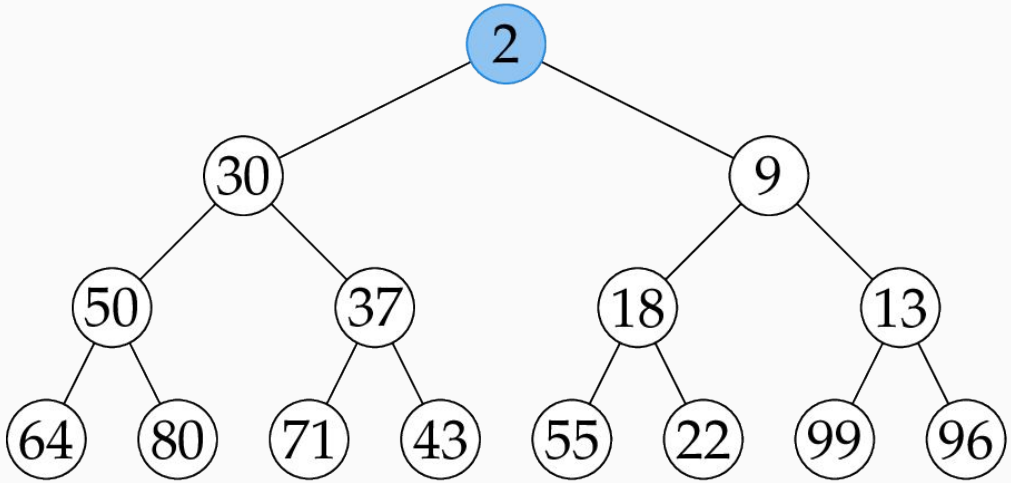

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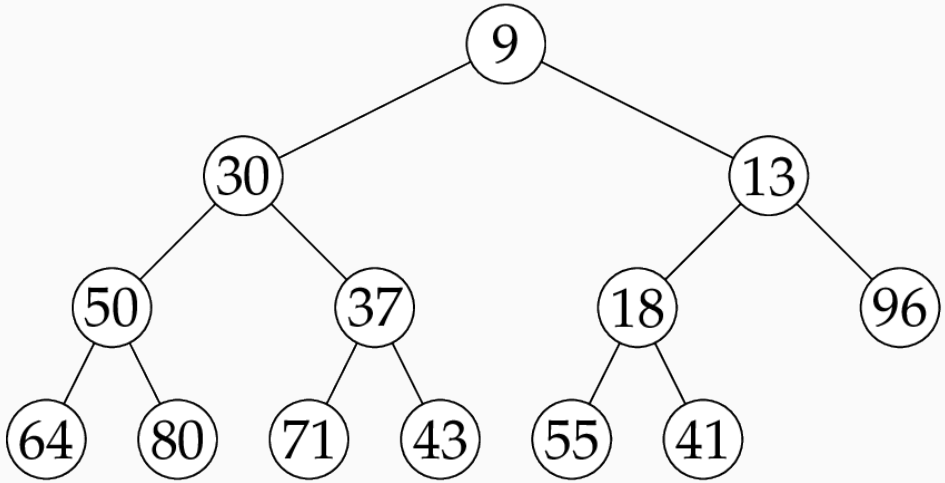

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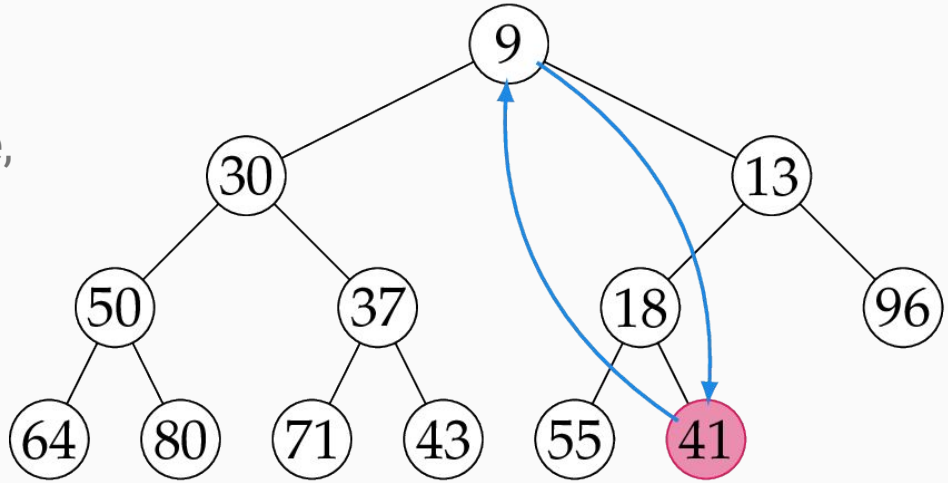

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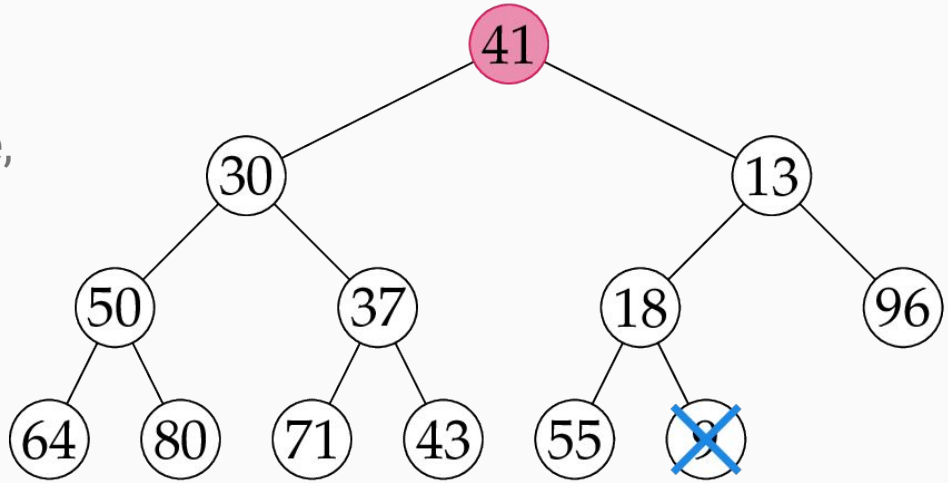

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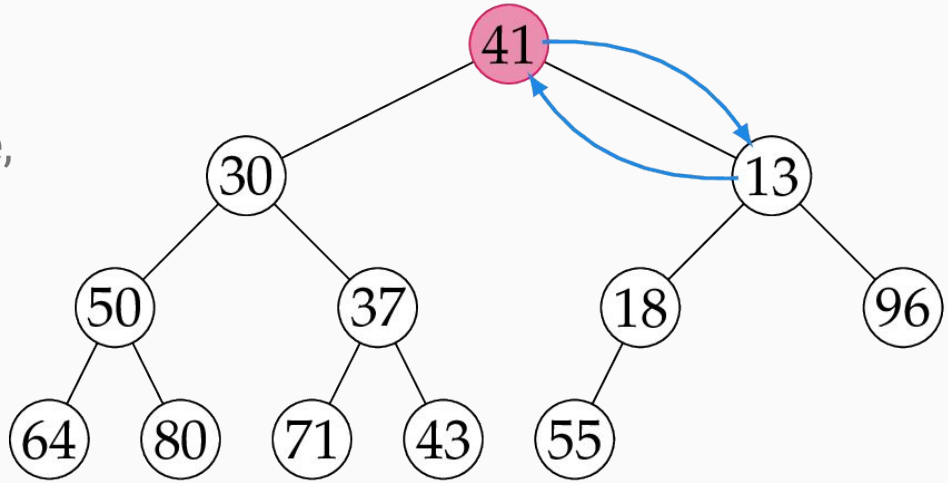

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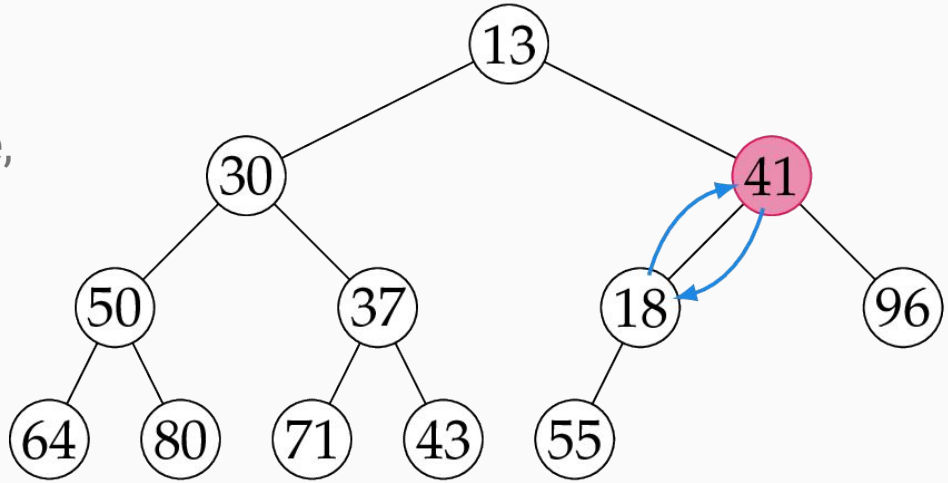

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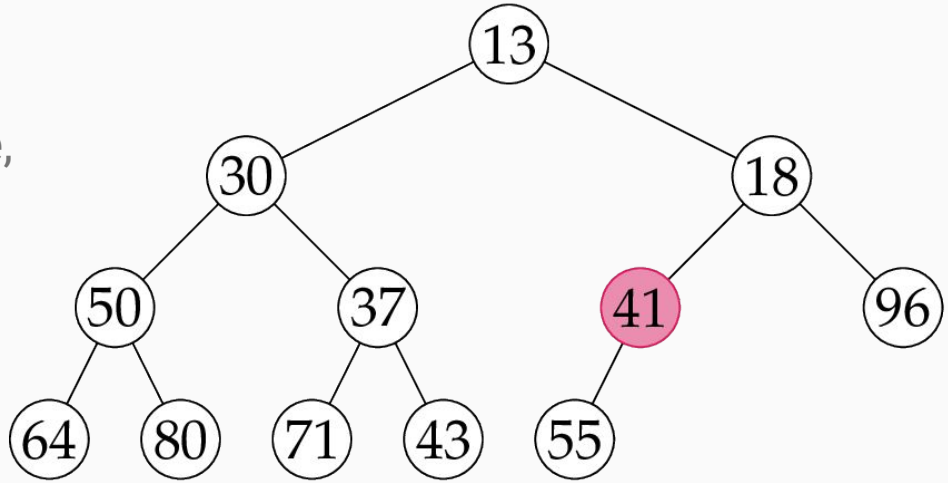

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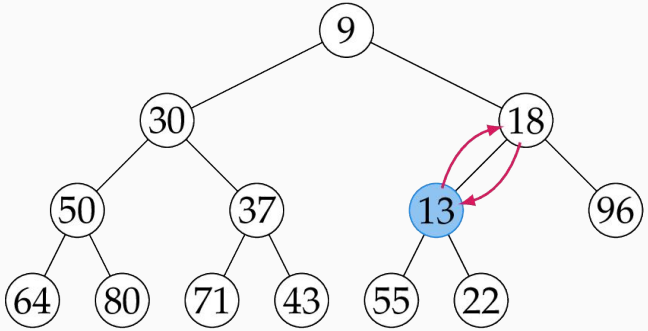

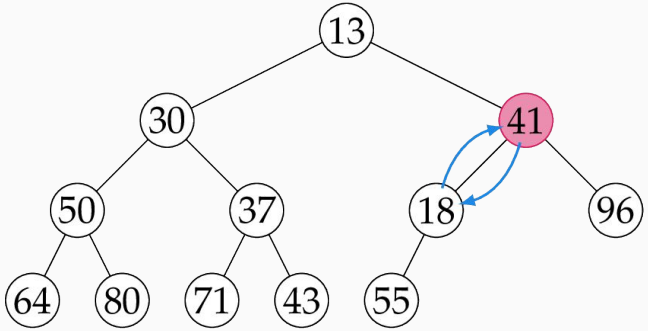

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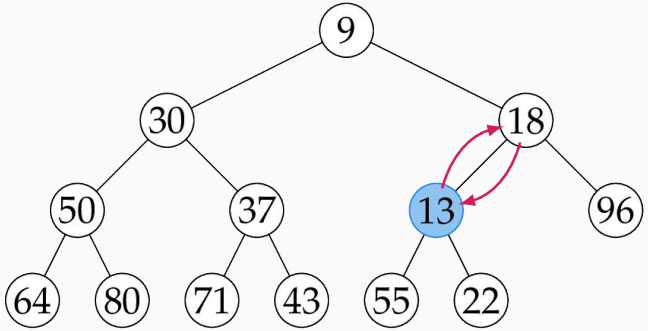
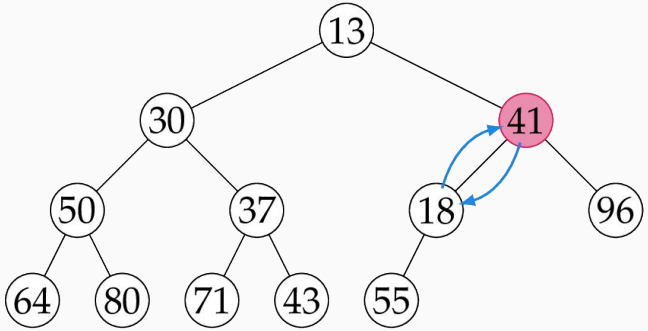
(better hidden constants than BST)





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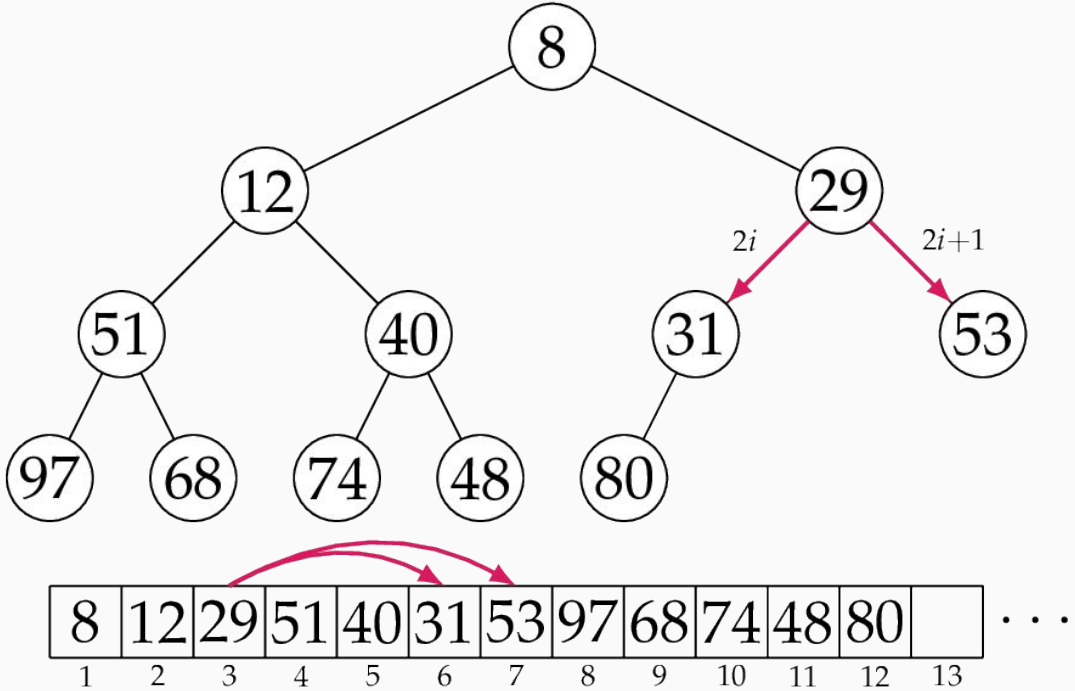

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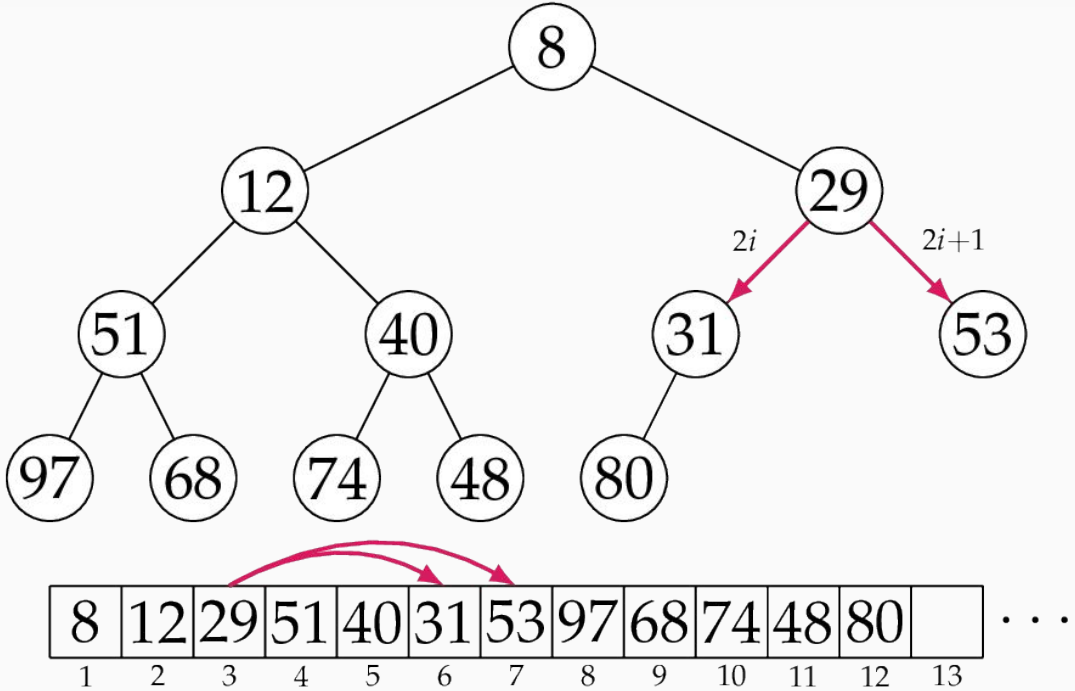

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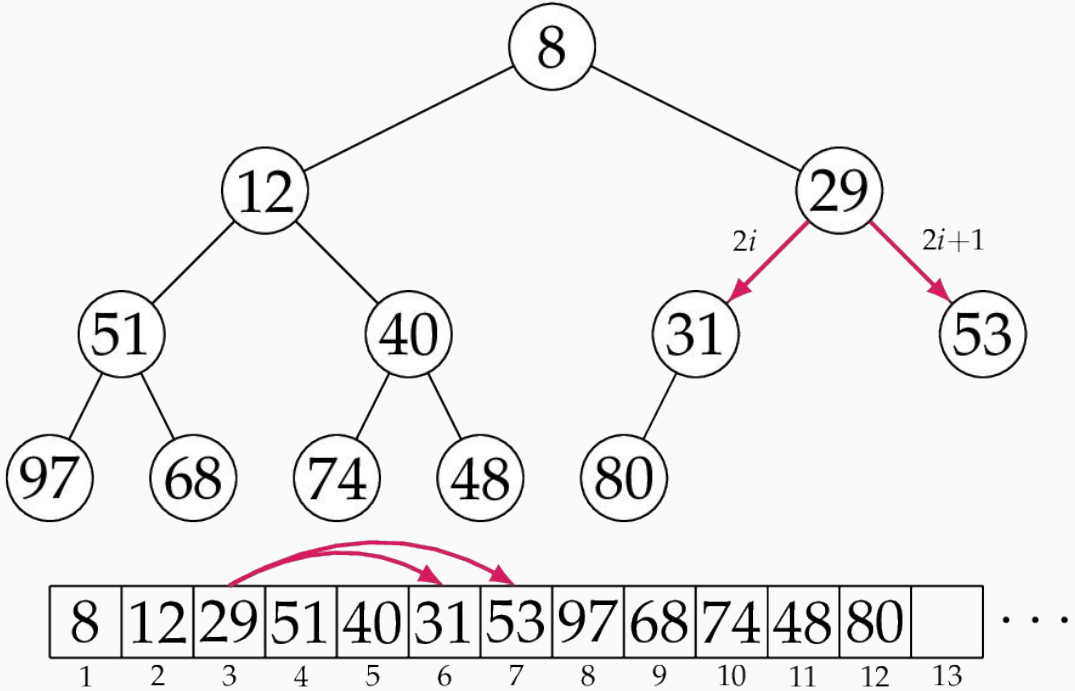
Why isn't this good for all binary trees?



Implementations

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… but these other approaches aren't entirely useless

Heapify

Building a heap from unsorted array in ***linear*** time:

- From bottom layer to top, bubble down

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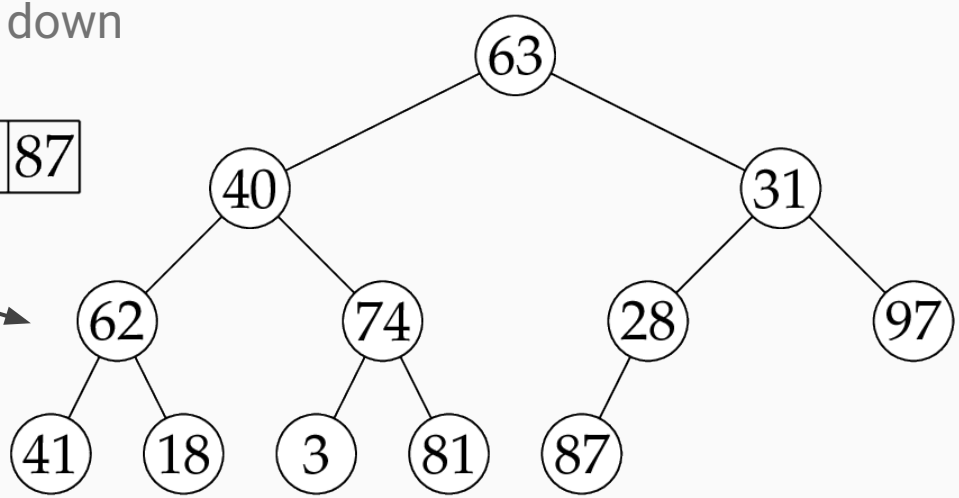

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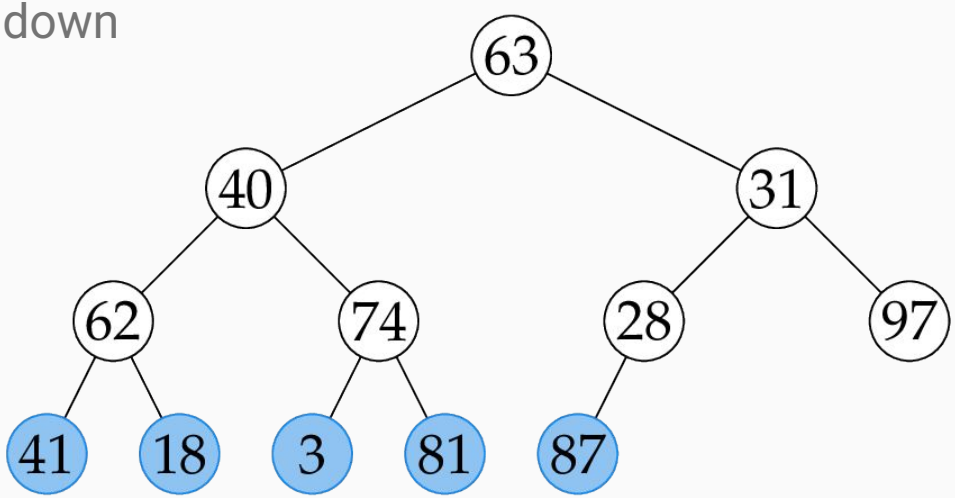

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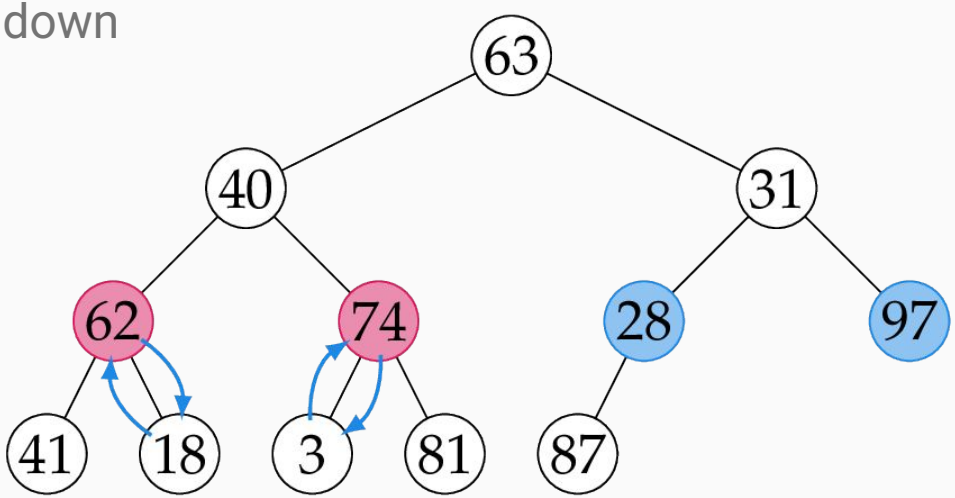

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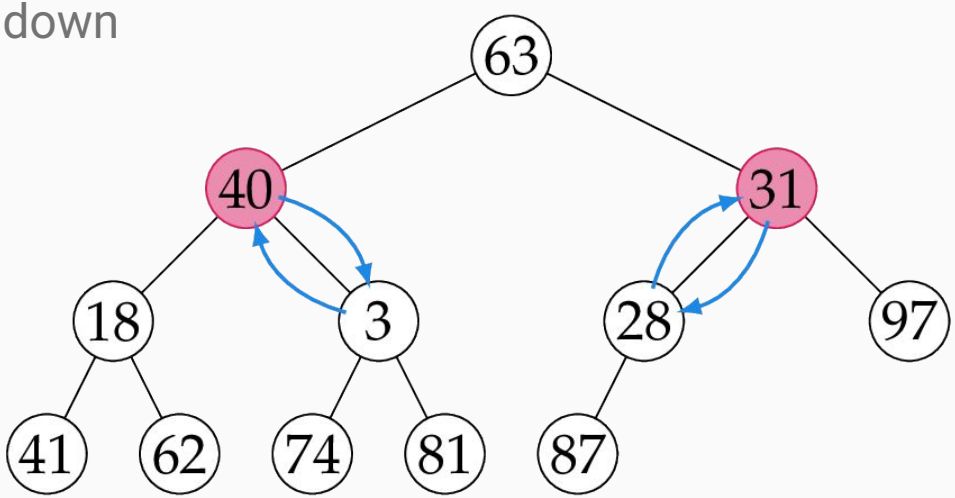

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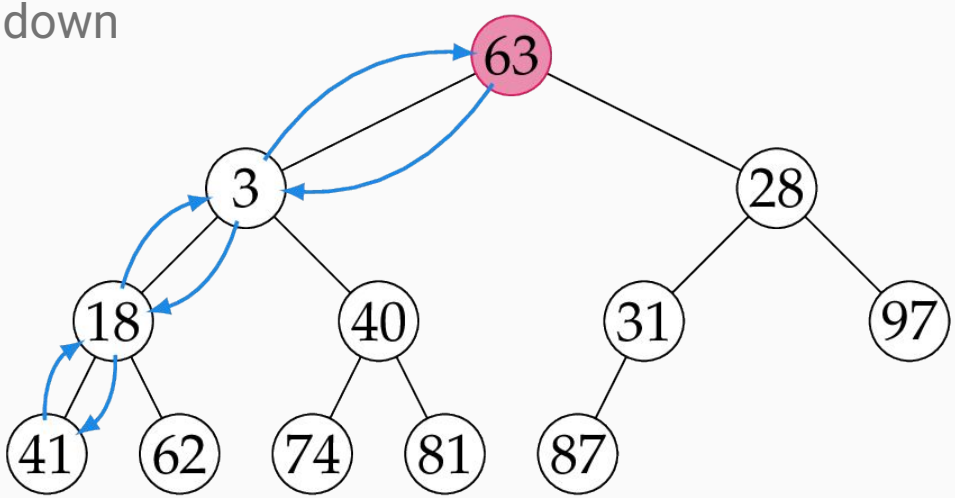

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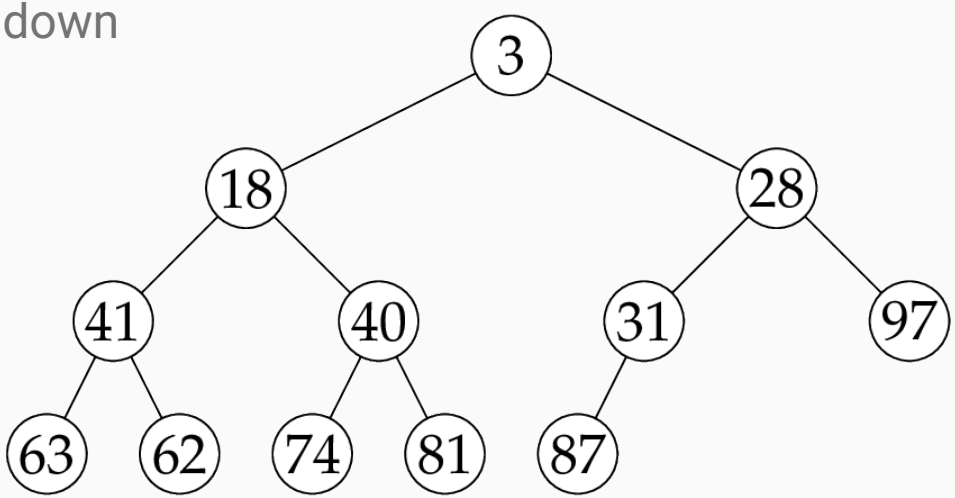

Heapify - correctness

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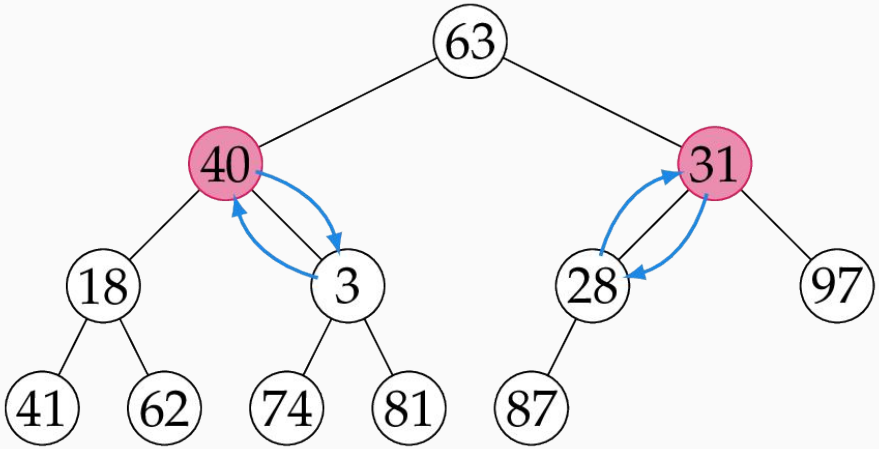

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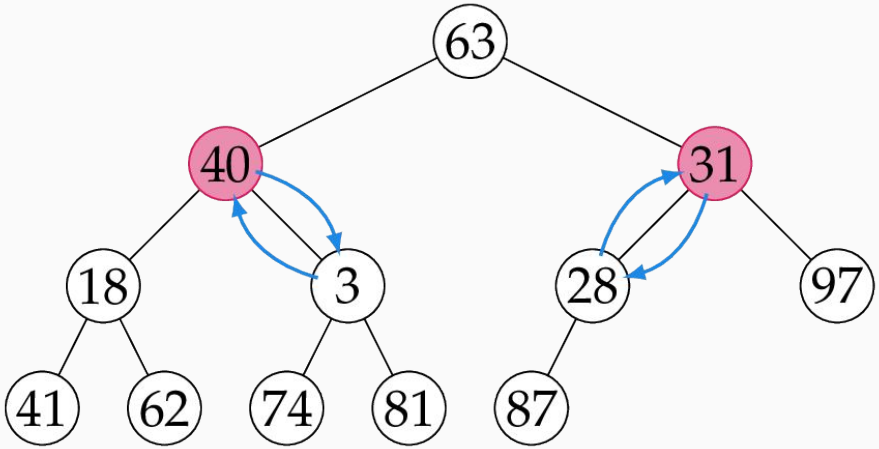
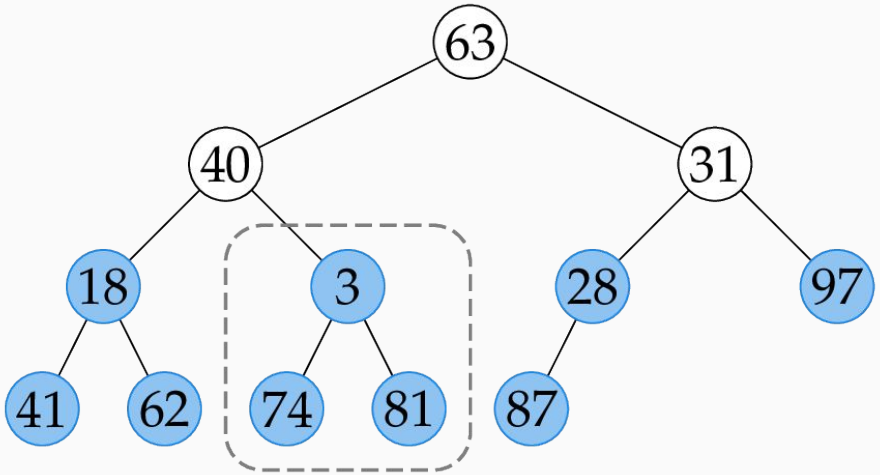
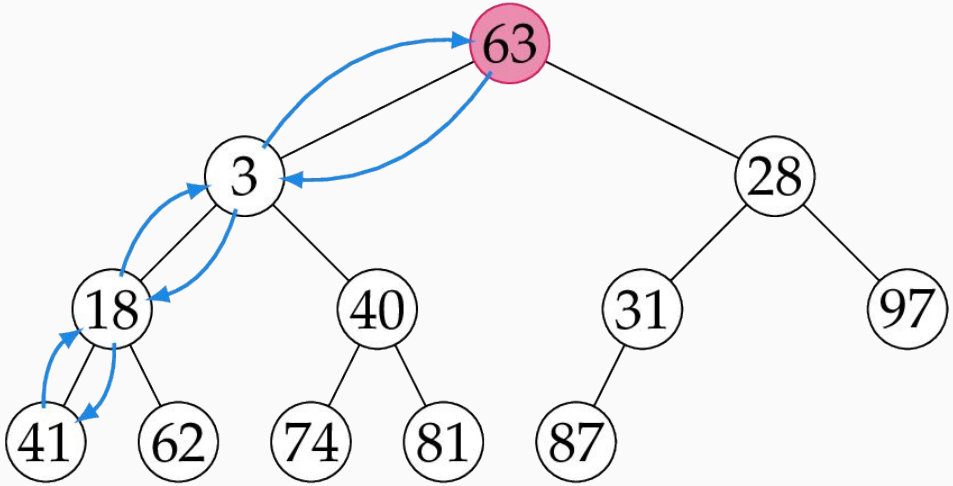




Heapify - runtime

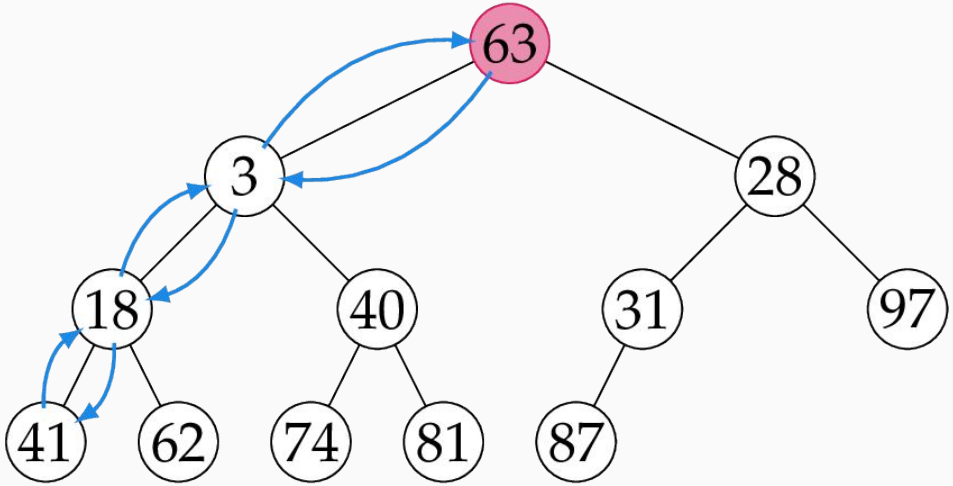
What’s the runtime?



Heapify - runtime

What's the runtime?

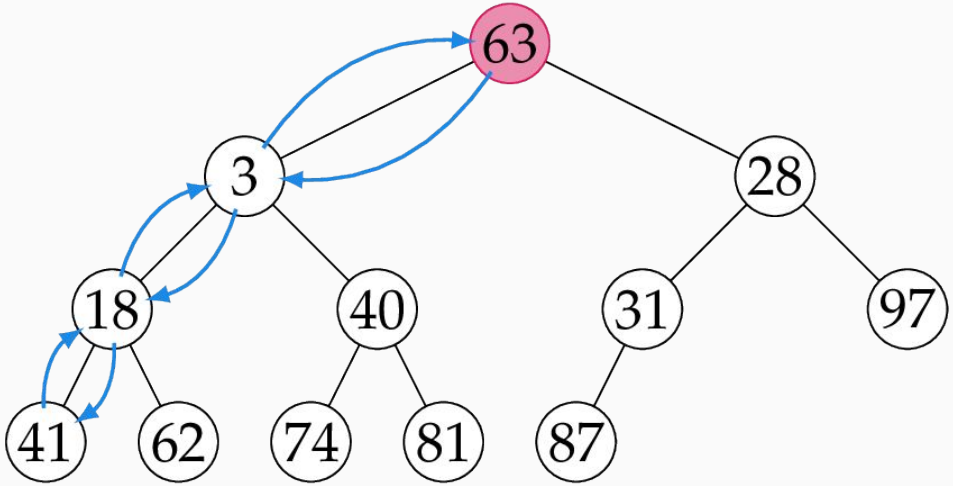
- ***n*** bubbling operations (O(log ***n***)?)



Heapify - runtime

What's the runtime?

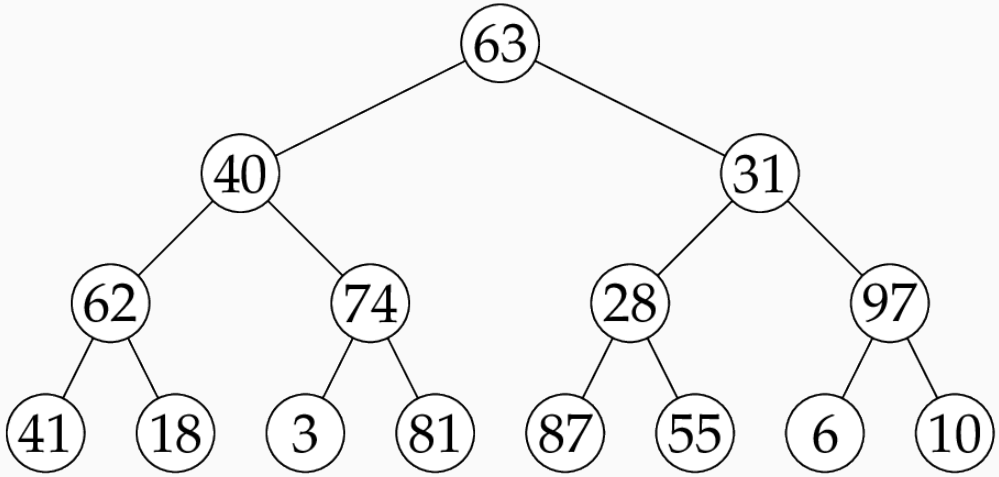
- ***n*** bubbling operations (O(log ***n***)?)
- O(***n*** log ***n***) time???



Heapify - runtime

Suppose bottom layer is full and consider, for each node, the following path:

1. Go to right child
2. Repeatedly go to the left child until you reach a leaf



Heapify - runtime

Suppose bottom layer is full and consider, for each node, the following path:

1. Go to right child
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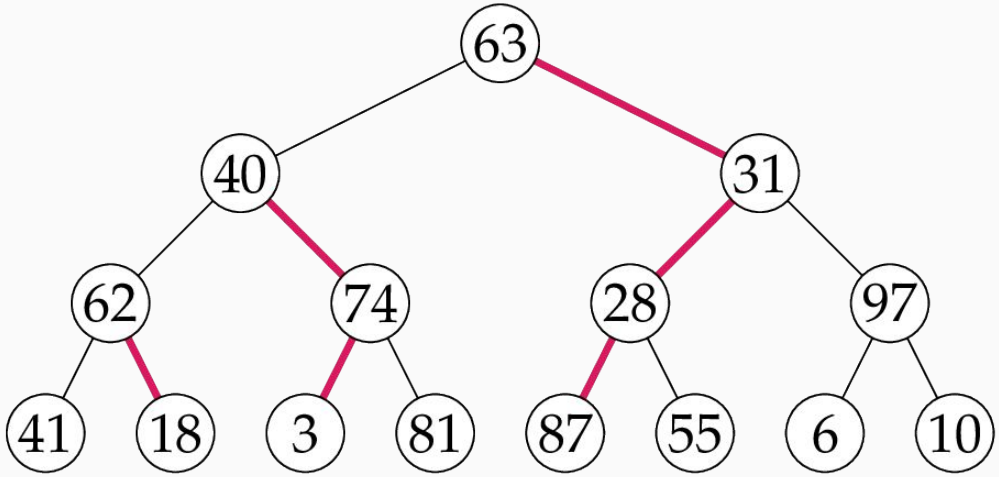
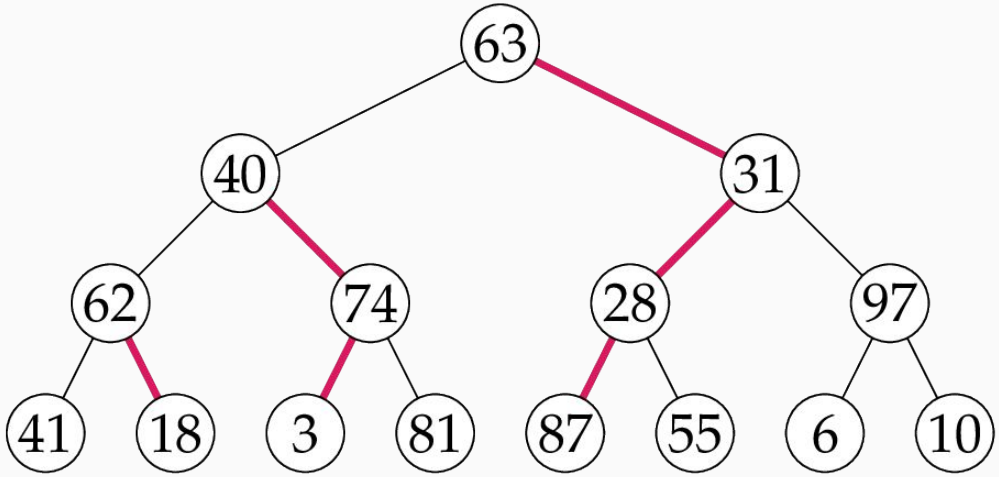
Each path is a potential “bubble-down”



Heapify - runtime

None of these paths overlap, so each edge is used at most once.



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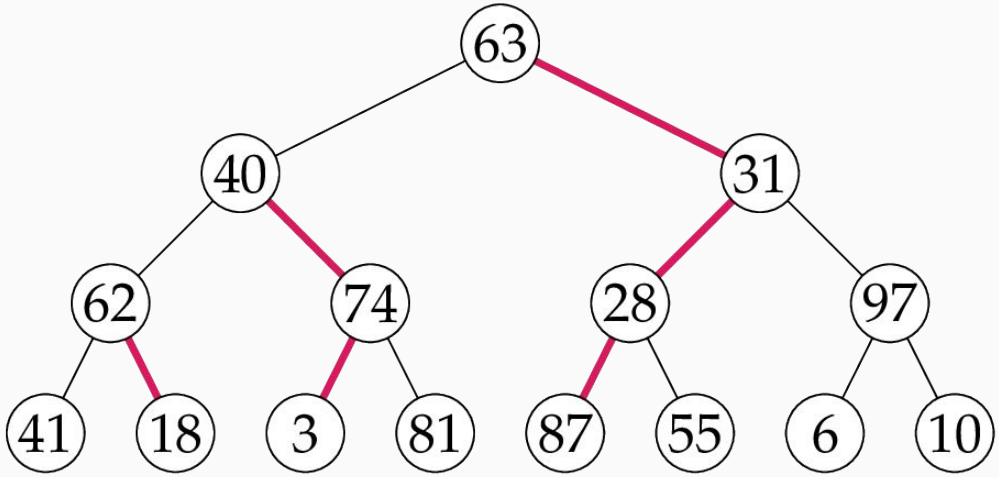
$|E| = n - 1$



Heapify - runtime

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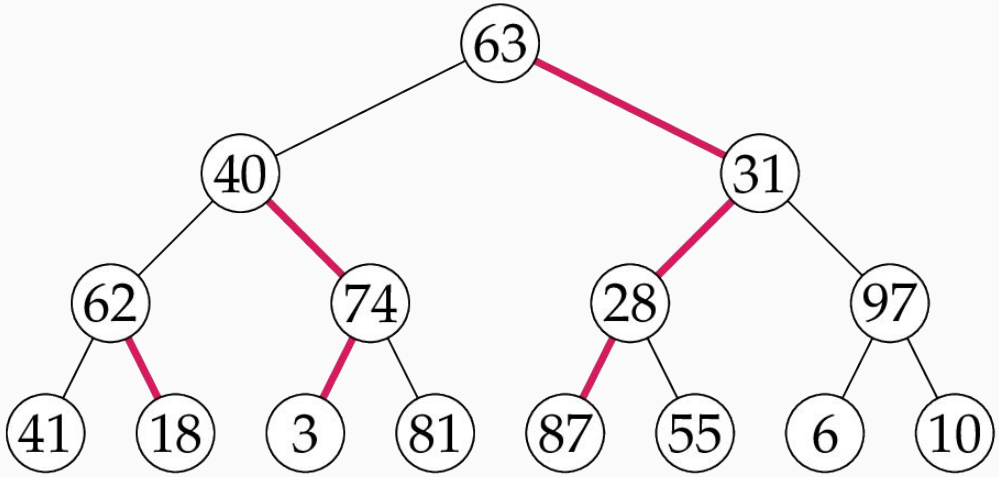
$$|E| = n - 1 \rightarrow O(n)$$



Heapsort

A simple sorting algorithm using a priority queue:

- Build a priority queue with all the elements inside.
- Repeatedly call removeMin( )

Heapsort

A simple sorting algorithm using a ***heap***:

- Build a ***heap*** with all the elements inside. → O(***n***)
- Repeatedly call removeMin( ) → ***n*** O(log ***n***) = O(***n*** log ***n***).

Total runtime: O(***n*** log ***n***)

“Extended” heap operations

In a binary heap, possible to modify other elements besides the root:

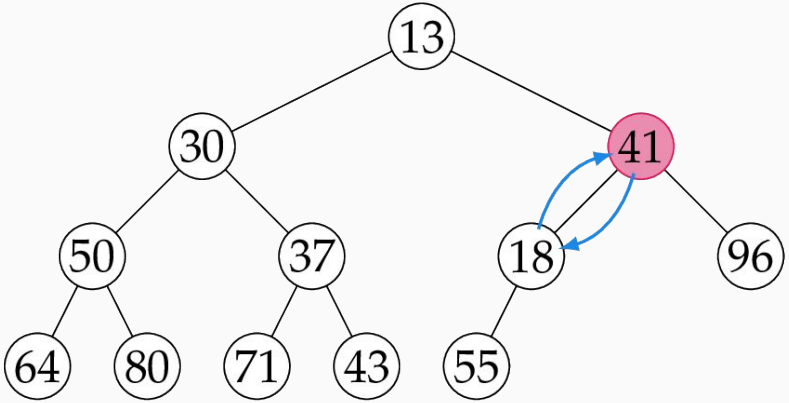
- decreaseKey(*key*, *newPriority*)
- remove(*key*)

“Extended” heap operations

In a binary heap, possible to modify other elements besides the root:

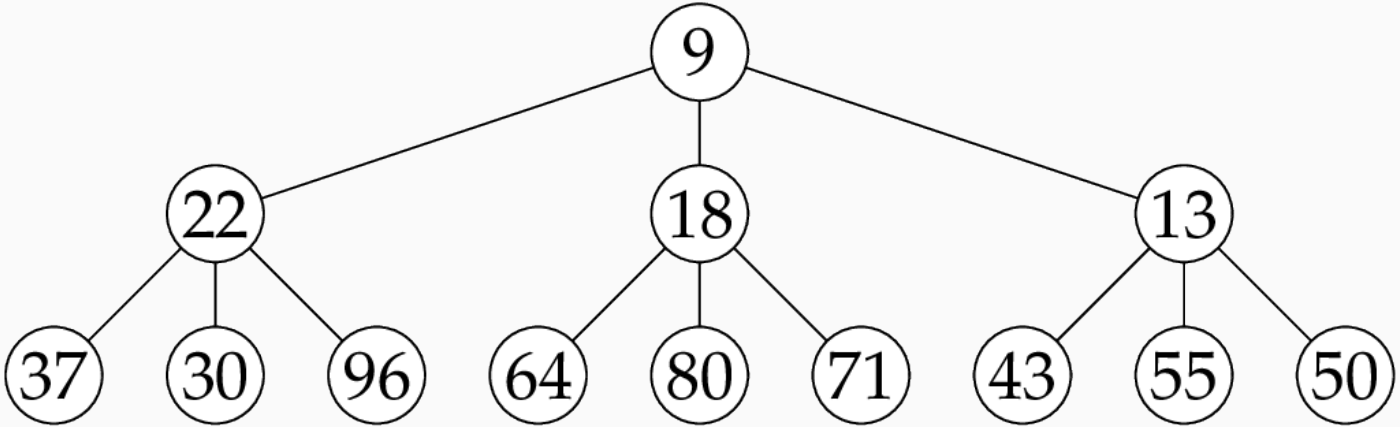
- decreaseKey(*key*, *newPriority*)
- remove(*key*)

Uses same bubbling operations as before



d-ary heaps

No reason why a heap can't have more than two children (what about BSTs?)



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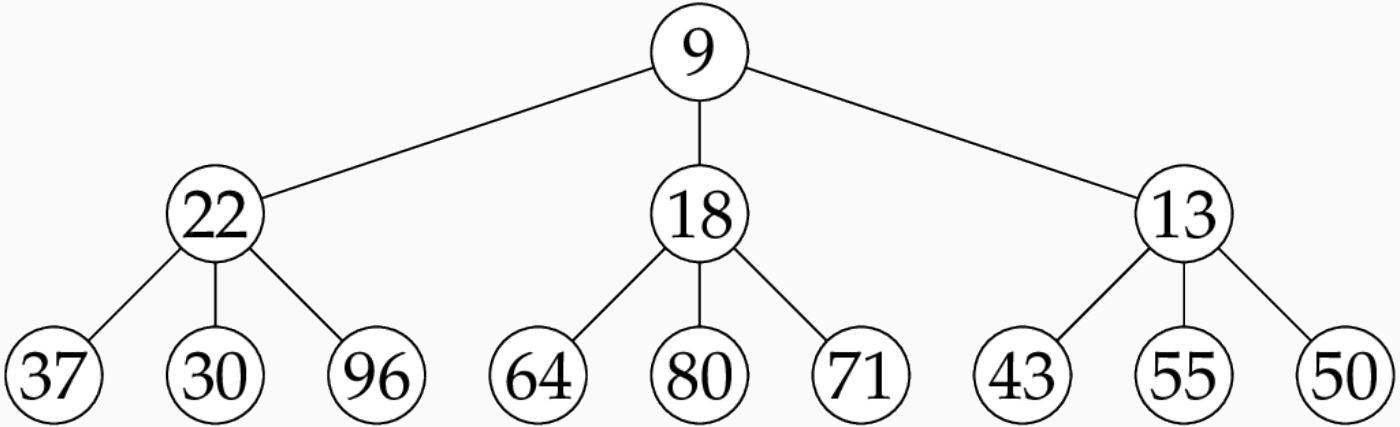
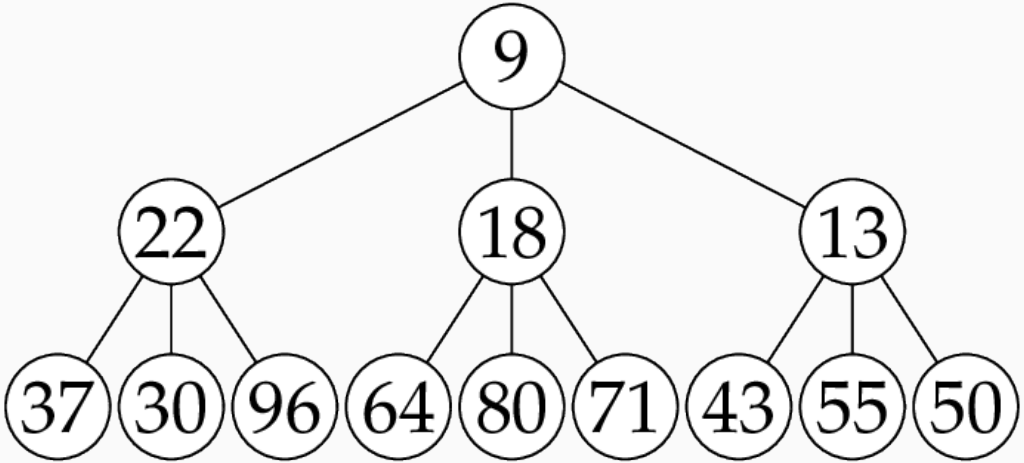


How does this affect runtime?

d-ary heaps

Still O(log ***n***), but maybe you care about hidden constants.

- Bubbling up is shorter.
- Bubbling down is longer.